

Ice Cream Tycoon: Shards & Payments Plan

Status: parked (nothing built yet). Written October 6, 2026.

Where things stand

Shards are **not built yet**. This document is the plan for adding shards (a premium currency bought with real money) and connecting them to Stripe for one-time purchases. The game itself keeps being built in the meantime; none of this blocks that work.

Important: the single-file test version (ice-cream-tycoon.html) can never take real payments because it has no server. Shards will only exist in the online game, and like the Cyber Event they can start out visible to admins only.

Two problems to fix first

- **Players don't have real accounts.** Today anyone can type any name and play as that person (admin names are the exception: they're locked to one device). With paid shards, someone could type a friend's name and spend their shards. So we need **accounts with a password**.
- **Progress can be lost.** Render's free server wipes its files on every update. Today progress comes back from a copy saved in each player's browser. That's fine for game money, but paid shards can't depend on it, so we need a **real database** (Postgres on Render).

The plan, in phases

Phase 1: Accounts and a database

Who: the developer builds it; you create the database.

- Sign up with a name and password, then log in. Existing players keep their progress when they sign up with their current name.
- All progress lives in a Postgres database on Render, so updates can't wipe it.

Phase 2: Shards and the Shard Shop (no real money yet)

Who: the developer builds it.

- A shard counter, and a **Shop button in the corner** that opens the Shard Shop.
- The shard packs (table below).
- The Game Passes can be bought with shards. Earn-only pets and lucky blocks can **not** be bought with shards (lucky blocks for real money would be a loot box). Shards can't be traded.
- A small free amount of shards each week from playing.
- Admins have unlimited shards and a **Gift Shards** button: a list of everyone playing pops up, tap a name, pick how many to give. Big gifts ask "Are you sure?" first, and every gift is logged.

Shard packs (the bonus is compared to 100 shards per \$1):

Price	Shards	Bonus
\$1	100	none
\$5	550	+10%
\$10	1,200	+20%
\$20	2,600	+30%
\$35	4,800	+37%
\$50	7,250	+45%
\$75	11,250	+50%
\$100	16,000	+60%

Phase 3: Stripe Checkout in Test mode

Who: the developer builds it; you add the keys.

- 1 The player taps a pack in the game.
- 2 The server asks Stripe for a checkout page for that pack and that account.
- 3 The player goes to **Stripe's own payment page**, so card numbers never touch the game.
- 4 When the payment goes through, Stripe tells the server directly (a "webhook"). Only then does the server add the shards. The "thank you" page alone never adds shards, so nobody can fake a purchase.
- 5 Each payment adds shards only once, even if Stripe sends the message twice.
- 6 If you refund a payment in Stripe, the shards are taken back automatically.

Phase 4: Keeping it safe and fair for kids

Who: the developer builds it.

- A **monthly spending limit per account, \$20 by default**. Only a parent can raise it.
- A parent email on the account, which gets a receipt for every purchase.
- A purchase history for each account, plus one for admins.

Phase 5: Go live

Who: mostly you.

- Swap the test keys for live keys, only after everything works in Test mode.

What you need to do, step by step

In Stripe (do this in Test mode, using the toggle at the top of the dashboard)

- 1 Go to **Developers** → **API keys** and copy the **Secret key**. In Test mode it starts with sk_test_.
- 2 After Phase 3 is built, go to **Developers** → **Webhooks** → **Add endpoint**:
URL: `https://ice-cream-tycoon.onrender.com/stripe/webhook` (or your Render address)

Events: `checkout.session.completed` and `charge.refunded`

Then copy the **Signing secret**, which starts with `whsec_`.

- 3 You don't need to create products in Stripe. The game sends each pack's price when the player checks out.

In Render

- 1 Go to **New** → **PostgreSQL** and create a database. Free databases expire after a while, so pick a cheap paid plan for anything that stores purchases.
- 2 Open the game's service, then **Environment**, and add:
DATABASE_URL = the database's **Internal Database URL**
STRIPE_SECRET_KEY = your `sk_test_...` key
STRIPE_WEBHOOK_SECRET = your `whsec_...` secret

Never paste these keys in a chat, in code, or on GitHub. They go only in Render's Environment settings.

Before switching to live money

- Test purchases with Stripe's test card 4242 4242 4242 4242 (any future date, any CVC). Also check that a refund takes the shards back.
- Stripe requires a page on the site with your **contact info, refund policy, and what you're selling**. This page can be built into the game's site.
- The person paying must be an adult cardholder. Since the players are kids, the parent email and spending limit matter. Laws about kids' games and data (like COPPA in the US) apply; it's worth checking with someone who knows before going live.

Decisions to make when you come back

- 1 **Accounts:** require a password for everyone, or only for players who want to buy shards? (Everyone is safer.)
- 2 **Monthly limit:** \$20 a month means the \$35 to \$100 packs can't be bought unless a parent raises the limit. Keep those packs, or drop them?
- 3 **The \$1 pack:** Stripe keeps about 2.9% plus 30 cents of every payment, so on \$1 about a third is lost. Keep it, or start at \$5?
- 4 **Free shards:** about 50 to 100 a week from playing. Still the plan?

To pick this back up: answer the four decisions above, create the Render database, and add DATABASE_URL. That's all Phase 1 needs; the Stripe keys come later.